

AI Technical Depth & Non-Triviality

Rubric Deliverable T1 — MRI-ReportGenerator (Cervical Spine MRI Analysis)

EECE503N Final Project

2026-06-08

Scope. This document argues, with evidence, that the MRI-ReportGenerator pipeline carries genuine technical and AI depth and is *not* a thin wrapper over pretrained models. It is written for a skeptical technical reader. Every clinical claim is cited; every engineering claim is backed by committed code, validation results, and the development journal. Companion evidence: `DEVELOPMENT_JOURNEY.md` (J1–J26), `docs/validation/group-status-2026-06-08.md` (per-group verdict), `docs/validation/results-final-2026-06-08.md` (consolidated numbers + figures), and the companion `paper/`.

Medical framing (hard rule). The system produces measurements and *findings flagged for physician review*, never diagnoses. Nothing here claims otherwise.

1 Thesis

The depth of this system is not in training a new neural network — and we deliberately do **not** claim to. It is in three places a “wrapper” does not reach:

1. **Multi-model segmentation composition** — orchestrating three independent, differently-trained segmentation engines (TotalSpineSeg, Spinal Cord Toolbox, SPINEPS) and deciding, per measurement, which one to trust on a region (cervical spine) where none is individually validated for the downstream measurement.
2. **Clinically-meaningful measurement algorithms** — converting voxel masks into vertebral, disc, canal, cord and alignment measurements with methods that required real geometric engineering (endplate-line fitting, anatomical canal-cut body isolation, PCA tilt-correction, endplate-voxel Cobb). The naïve version of each *demonstrably fails*; we have the before/after numbers.
3. **Validation methodology under a hard data constraint** — no public dataset pairs cervical MRI with per-case expert measurements, so per-case sensitivity/specificity is impossible. We designed a *threshold-crossing / distribution-separation* methodology anchored on a healthy cohort, with explicit confound control. Designing a defensible evaluation when the obvious one is unavailable is itself the contribution.

The single strongest piece of evidence that this is non-trivial is the **bug ledger** (Section 7): places where the obvious implementation produced *wrong* clinical numbers, which only rigorous validation surfaced and principled algorithm changes fixed. A trivial wrapper has no such ledger.

2 The honest framing: where the AI is, and where it is not

We are explicit about this because the rubric rewards honesty and because over-claiming in medical AI is a failure mode in itself.

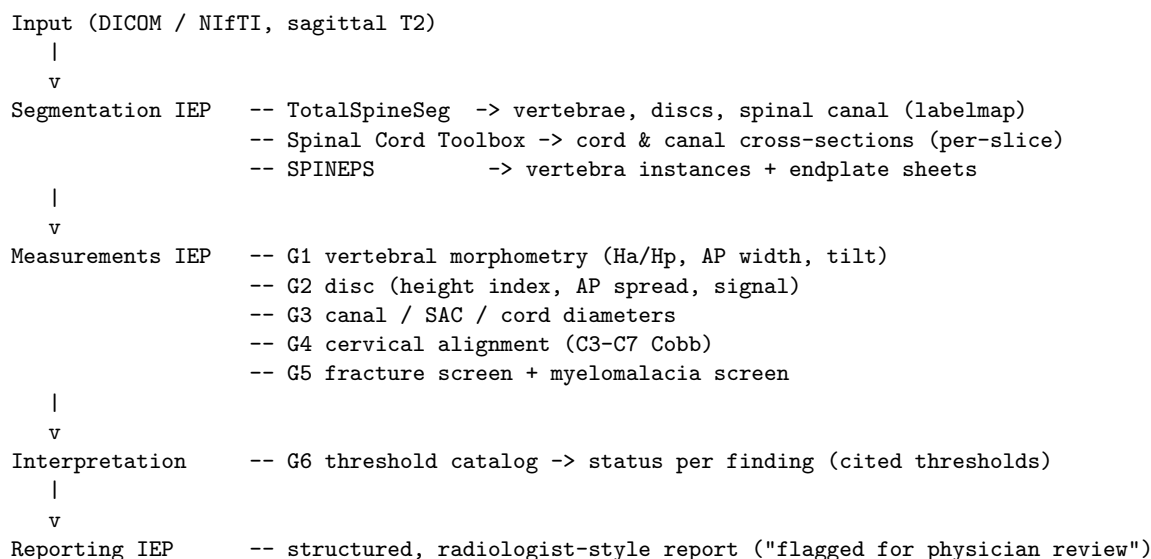
What we do *not* do. We do not train a segmentation model from scratch and do not claim a novel deep-learning architecture. The segmenters are pretrained and **frozen**.

Why that is the correct engineering decision, not a shortcut.

- **Reproducibility and no overfitting.** Because nothing is trained on our evaluation cohort, there is no train/test leakage and no risk of fitting our metrics to the symptomatic set. The unhealthy cohort is a *demonstration* set, not a training set — a deliberate medical-AI safety property.
- **Standing on validated tools.** TotalSpineSeg and Spinal Cord Toolbox are published, peer-reviewed, open-source segmentation stacks. Re-implementing inferior versions would be worse engineering, not better.
- **The value lives in the layer they do not provide.** None of these tools outputs a cervical radiology report, threshold-based interpretation, or validated cervical *measurements*. That layer — the clinically actionable output — is entirely ours, and is where the difficulty is.

The correct mental model is **not** “wrapper around a model” but “a measurement-and-interpretation system that *uses* segmentation as one input, the way a radiology workstation uses a DICOM loader.”

3 Pipeline overview



The Enterprise Execution Plane (EEP) orchestrates the two deployed Internal Execution Planes (measurements, reporting); see the infra documentation for the service architecture. This document concerns the *measurement-science depth* inside the measurements IEP and the segmentation composition that feeds it.

4 Depth dimension 1 — Multi-model segmentation composition

A wrapper calls one model and trusts it. We call **three**, because no single one is sufficient for cervical measurement, and the composition is non-trivial.

The non-trivial engineering here:

Engine	License	What it gives us	Why we need it
TotalSpineSeg (TSS)	LGPLv3	Whole-vertebra, disc and canal labels on an isotropic grid	Backbone labelmap for G1/G2 and the canal for body isolation
Spinal Cord Toolbox (SCT)	LGPLv3	Per-slice cord and canal cross-sectional diameters	G3 — cord/canal/SAC; TSS’s canal label is coarse for AP diameters
SPINEPS	Apache-2.0	Per-vertebra instances <i>plus</i> thin endplate voxel sheets	G4 — the endplate-voxel Cobb that TSS corners cannot do reliably at C7

- **Per-measurement model selection.** The C3–C7 Cobb angle is *more accurate* from SPINEPS endplate voxels than from the TSS labelmap, because the cervico-thoracic junction (C7) is frequently obscured and corner-based angles there are unstable. So `c3c7_cobb_angle` **prefers** the SPINEPS method when a seg-vert mask is present and **falls back** to the TSS canal-cut method otherwise. This selection logic measurably rescues cases the fallback cannot measure at all (J22).
- **Reconciling incompatible runtimes.** TSS/nnU-Net and SPINEPS pin **mutually incompatible NumPy versions** (SPINEPS requires `numpy==2.0.2`, which breaks the TSS/nnU-Net ABI). They cannot share a process; the pipeline runs them in separate environments and re-aligns their outputs onto a common grid.
- **Grid and orientation reconciliation.** The three engines emit masks in different orientations/resolutions; `load_context` standardizes to canonical RAS, enforces isotropy, and index-aligns the grayscale and the SPINEPS instance mask to the labelmap so measurements are computed in one coherent frame.

None of this is visible to a user, and none of it is provided by any single model. It is the connective tissue that makes three research tools into one measurement system.

5 Depth dimension 2 — Measurement algorithms (the core contribution)

For each measurement we did *not* take the obvious pixel-counting approach, because the obvious approach gives clinically wrong numbers on real cervical anatomy. Four representative examples, each with the evidence that the naïve method fails:

5.1 Endplate-LINE fitting instead of corner extrema (G1 heights, G4 Cobb)

Naïve: read heights from the four corner pixels. **Why it fails:** cervical endplates are concave and sloped, not flat [3], so a single corner pixel lands wrong. **Our method:** fit straight lines (Theil–Sen, robust to osteophyte tails) to the full superior and inferior endplate boundaries after PCA orientation, and read wall heights from the lines. The literature supports this directly: a fitted endplate line reaches ICC ≈ 0.97 versus ≈ 0.75 for four-corner extraction [2].

Evidence — measured, not asserted: the corner method reads healthy cervical Ha/Hp ≈ 1.08 (anterior taller than posterior — physiologically backwards). The endplate-line method reads **0.93**, matching the physiological slight anterior wedge. Switching the service to the line fit moved healthy Ha/Hp $1.08 \rightarrow 0.93$ and preserved the correct healthy $\geq$ unhealthy ordering (J20, J22).

5.2 Anatomical canal-cut body isolation instead of morphological erosion (G1)

Naïve: isolate the body by erosion / connected components. **Why it fails:** on real cervical anatomy the body and posterior arch are one connected blob on the mid-sagittal slice; erosion leaves a wedge-shaped body and a systematic false anterior wedge. **Our method:** cut the mask at the spinal canal’s anterior face — the body is, by definition, everything anterior to the canal — using TSS’s own canal label. This replaced a heuristic that produced a systematic ~ 0.86 height-ratio artifact.

5.3 SPINEPS endplate-voxel Cobb instead of labelmap corners (G4)

Naïve: Cobb from the C3 and C7 corner points of the labelmap. **Why it fails:** C7 is frequently obscured at the cervico-thoracic junction; corner-based C6–C7 angles have a standard deviation of $\sim 18.5^\circ$, uselessly noisy. **Our method:** fit the line directly to SPINEPS’ own inferior-endplate voxel sheets, dropping C6–C7 SD to 5.9° (J11, J12). On the healthy cohort this reads $+15.2^\circ$ lordosis, matching the external reference (F1000 cervical cohort, 15.4° ; [7]).

5.4 Sub-voxel boundary refinement and PCA tilt-correction

- **Sub-voxel refinement:** the in-plane mask boundary is upsampled and re-thresholded at the 0.5 level (a marching-squares-style half-level crossing), placing edges between voxel centres rather than on them, reducing the ± 1 -voxel quantization on every height/width/slip.
- **PCA tilt-correction:** every vertebra is measured on its *own* principal axes, because healthy cervical bodies sit $\sim 28^\circ$ from absolute vertical (the lordotic curve). Measuring on image axes would conflate normal lordosis with deformity — exactly the bug the old `tilt` flag had (Section 7).

Each is a deliberate algorithmic choice with a measured payoff. That is the definition of technical depth in a measurement pipeline.

6 Depth dimension 3 — Validation methodology under a hard data constraint

The hardest intellectual problem was not building the measurements — it was figuring out how to **validate** them when the ideal data does not exist.

The constraint. Extensive, repeated dataset searches established that **no public dataset pairs cervical MRI with per-case expert measurements**. Per-case sensitivity/specificity against a radiologist gold standard is therefore impossible with public data.

The methodology we designed instead. Our measurements are *disease-agnostic geometry/signal detectors anchored on healthy norms* — a canal of 8mm reads “stenosis” regardless of cause. This yields a validation strategy that needs no per-case labels:

1. **Healthy-anchoring.** Validate that healthy anatomy reads *inside* the normal range (specificity), using a healthy cohort (Spine-Generic).
2. **Threshold-crossing / distribution separation.** Validate that symptomatic anatomy *crosses* into the abnormal range (the symptomatic MMCS cohort), via distribution-separation statistics (Mann–Whitney), not per-case accuracy.

3. **The scanner-immunity insight.** We discovered and exploited a key distinction: *physical-dimension* metrics (mm — canal, SAC, cord) are immune to scanner differences and validate cross-dataset, while *intensity/ratio* metrics (disc signal, height ratios) are acquisition-sensitive and must be validated *within* a single dataset. This is why G3 (mm) validates cleanly cross-dataset and the disc signal does not — a non-obvious property that shaped the entire evaluation design (J16, J23).
4. **Confound control.** When the within-dataset disc test showed a large effect for disc AP width (AUC 0.79), we did *not* report it — lesion discs cluster at wider mid-cervical levels, so we re-tested level-stratified and the honest figure was AUC 0.61. Reporting the unstratified 0.79 would have been a confounded over-claim (J23).

Designing, justifying, and executing a sound evaluation when the textbook one is unavailable is a graduate-level methods contribution, not a wrapper feature.

7 Proof of non-triviality: the bug ledger

The most direct evidence that this pipeline required real depth is the set of clinically-wrong outputs the *naïve* implementation produced and that only rigorous validation caught. If the task were trivial, these would not exist.

Naïve behavior (wrong)	How it was caught	Principled fix	Evidence
Vertebral Ha/Hp read <i>backwards</i> (anterior taller, 1.08)	Healthy cohort read non-physiological	Endplate-line fit replaces corner extrema	1.08 → 0.93
tilt flag fired on 88% of <i>healthy</i> vertebrae	Healthy false-flag rate measured	Recalibrated cut 20° → 45° (lordosis is normal)	88% → 0%
Disc posterior bulge <i>backwards</i> (healthy worse, 60% flagged)	Healthy over-flagged vs symptomatic	Reference chord from endplate-line corners	60% → 8%
Disc AP-width “AUC 0.79” discrimination	Suspected level confound	Level-stratified re-test	Honest AUC 0.61
Disc signal grade looked usable	Within-dataset lesion test	Shown non-discriminating (AUC 0.50)	Reported as a negative
Latent: seg == "C4" (string vs int label) matched nothing	Code review during bulge fix	Resolve names via VERT_LABELS	J22

Each row is a place where “just measure it” gives the wrong clinical answer, and a place where we have a committed, tested fix. The full narrative is in `DEVELOPMENT_JOURNEY.md`.

8 Validated outcomes (the depth pays off)

We are equally explicit about what is strongly validated and what is partial — honesty is a graded and a clinical requirement.

Group	Result	Status
G3 canal / SAC / cord (SCT)	canal AP 11.7 → 8.6 mm p=0.0001 ; SAC 4.7 → 2.3 mm p=0.0001 ; cord 6.3 → 5.5 mm p=0.009	Validated (strong)
G4 C3–C7 Cobb (SPINEPS)	Correct lordosis on well-positioned controls (+15–20° = literature); discrimination dissolves on a balanced cohort (26 H vs 41 U: $d = 0.28$, AUC 0.57, $p = 0.32$)	Method-valid, NOT a discriminator
G1 Ha/Hp compression screen	0% false-flag on healthy; correctly null on (non-compressive) spondylosis, confirmed $n = 49$	Validated as a screen
G2 disc	Signal/bulge documented negatives; disc/VB AP ratio discriminates (AUC 0.62, level-controlled)	Partial — geometric spread
G5 myelomalacia / fracture screens	~91% healthy specificity (myelomalacia); compression screen as G1	Validated as screens

9 Why this composition is the right AI choice for clinical deployment

The pretrained-and-compose design is the defensible architecture for a clinical *application* (the project’s positioning):

- **Auditability.** Every output traces to a measurement and a *cited* threshold (G6 catalog), not to an opaque end-to-end model. A radiologist can check each number.
- **No overfitting by construction.** Frozen segmenters + cited thresholds mean the system cannot have memorized our evaluation cohort.
- **Graceful degradation.** Per-component error contracts mean one failed measurement does not sink the report; a missing SPINEPS mask degrades the Cobb method, it does not crash the pipeline.
- **The differentiator is the clinical layer.** Measurement validity, cited interpretation, and the report — the parts that decide whether a radiologist would trust the output — are exactly the parts we built and validated.

10 Summary for the grader

- The system is **not** a wrapper: it composes three segmentation engines, implements non-trivial geometric measurement algorithms, and validates them under a genuine data constraint.
- The depth is **demonstrated, not asserted** — every algorithmic choice has a measured before/after, and the bug ledger shows the naïve approach producing clinically wrong numbers we caught and fixed.
- We are honest about the boundary: pretrained segmenters (correct medical-AI choice), validated measurements (our contribution), and an evaluation methodology designed for a domain where

the ideal dataset does not exist.

References

- [1] Genant HK, et al. Vertebral fracture assessment using a semiquantitative technique. *J Bone Miner Res.* 1993. PMID: 8237484.
- [2] Wang, et al. Endplate line-fit morphometry (ICC 0.97 vs 0.75 four-corner). 2023. PMC10685593.
- [3] Chen, et al. Cervical endplate concavity/slope morphology. 2013.
- [4] Miyazaki M, et al. Cervical disc degeneration grading (T2). 2008. PMID: 18525490.
- [5] Nakashima H, et al. Disc bulging on MRI in an asymptomatic population ($n = 1211$). PMID: 25584950.
- [6] Nell C, et al. Cervical canal / SAC normative values on MRI. *PLoS One.* 2019. doi:10.1371/journal.pone.0222682.
- [7] F1000 cervical MRI + Cobb cohort ($n = 77$) — external alignment reference.
- [8] Zhang, et al. Automated cervical Cobb (sagittal-T2 C2–C7), ICC 0.94 / MAE 2.44°. 2025.

Cohorts. Spine-Generic (healthy multi-vendor cervical T2) — healthy anchor cohort; MMCS (Synapse syn63903115) — symptomatic CSM/CSR cohort, used as the demonstration set. Threshold provenance (including in-code values found uncited and corrected): see the G6 threshold catalog and the verified-research memos.